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Spectra-temporal patterns underlying mental addition: An ERP and ERD/ERS study

Yixuan Ku*, Bo Hong, Xiaorong Gao, Shangkai Gao

Department of Biomedical Engineering, Medical School, Tsinghua University, 1 Qinghuayuan, Haidian District, Beijing 100084, China

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ABSTRACT

Functional neuroimaging data have shown that mental calculation involves fronto-parietal areas that are composed of different subsystems shared with other cognitive functions such as working memory and language. Event-related potential (ERP) analysis has also indicated sequential information changes during the calculation process. However, little is known about the dynamic properties of oscillatory networks in this process. In the present study, we applied both ERP and event-related (de-)synchronization (ERS/ERD) analyses to EEG data recorded from normal human subjects performing tasks for sequential visual/auditory mental addition. Results in the study indicate that the late positive components (LPCs) can be decomposed into two separate parts. The earlier element LPC1 (around 360 ms) reflects the computing attribute and is more prominent in calculation tasks. The later element LPC2 (around 590 ms) indicates an effect of number size and appears larger only in a more complex 2-digit addition task. The theta ERS and alpha ERD show modality-independent frontal and parietal differential patterns between the mental addition and control groups, and discrepancies are noted in the beta ERD between the 2-digit and 1-digit mental addition groups. The 2-digit addition (both visual and auditory) results in similar beta ERD patterns to the auditory control, which may indicate a reliance on auditory-related resources in mental arithmetic, especially with increasing task difficulty. These results coincide with the theory of simple calculation relying on the visuospatial process and complex calculation depending on the phonological process.

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Mental calculation is an important capability for human beings. Functional magnetic resonance imaging (fMRI) research and neuropsychological studies have revealed that a range of fronto-parietal brain areas, together with the temporal lobe, play a role in arithmetic processing [4,24]. Electroencephalography (EEG) studies also revealed the neural correlates underlying mental calculation. Event-related potential (ERP) studies have shown sequential cognitive processes during mental arithmetic [9]. ERP is considered to be the phase resetting of ongoing EEG oscillations evoked by external stimuli [22]. However, some late cognitive brain oscillations are not ideally phase-locked, and a waveform may thus be eliminated in ERP averages, whereas event-related (de-)synchronization (ERS/ERD) analysis can preserve the dynamic, multi-band information in the frequency domain. There have been investigations on the spectral changes during mental arithmetic, which predominantly reveal an increase in frontal theta activity and a decrease in parietal alpha activity [8].

In clinical studies, patients with brain lesions have shown dissociation of Arabic and verbal number production, in addition to calculation [13]. However, an fMRI study has revealed that there may be a supra-modal calculation system for abacus mental calculation experts [3]. In the present study, we hypothesized that the non-modality specific calculation system also existed for ordinary people and designed both visual and auditory experiments to test this hypothesis. A non-phase-locked method was further applied to the auditory data because the spoken numbers could not be presented as briefly as the visual ones. Herein we used a joint method with both ERP and ERD/ERS analyses to explore the spectra-temporal patterns underlying cognitive processes during mental addition in the two modalities.

Fifteen right-handed college students were recruited for the study (11 males, 4 females, aged 21–27 years, mean ± SD = 23.6 ± 1.7 years). Participants signed informed consent and were compensated for their time. The protocols of the experiments were approved by the Institutional Ethics Committee of School of Medicine, Tsinghua University. The experiments were conducted in a dark, sound-proof room free of ambient electromagnetic interference. Participants were placed in a comfortable chair facing an LED screen at eye level with two speakers on either side of the screen while performing the tasks.

Six different tasks (V-2, V-1, V-L, A-2, A-1, and A-L) were used in this study. Three of them (V-2, V-1, and V-L) were visual tasks,

* Corresponding author. Tel.: +86 15001390906; fax: +86 1062792416.
E-mail address: yixuanku@gmail.com (Y. Ku).

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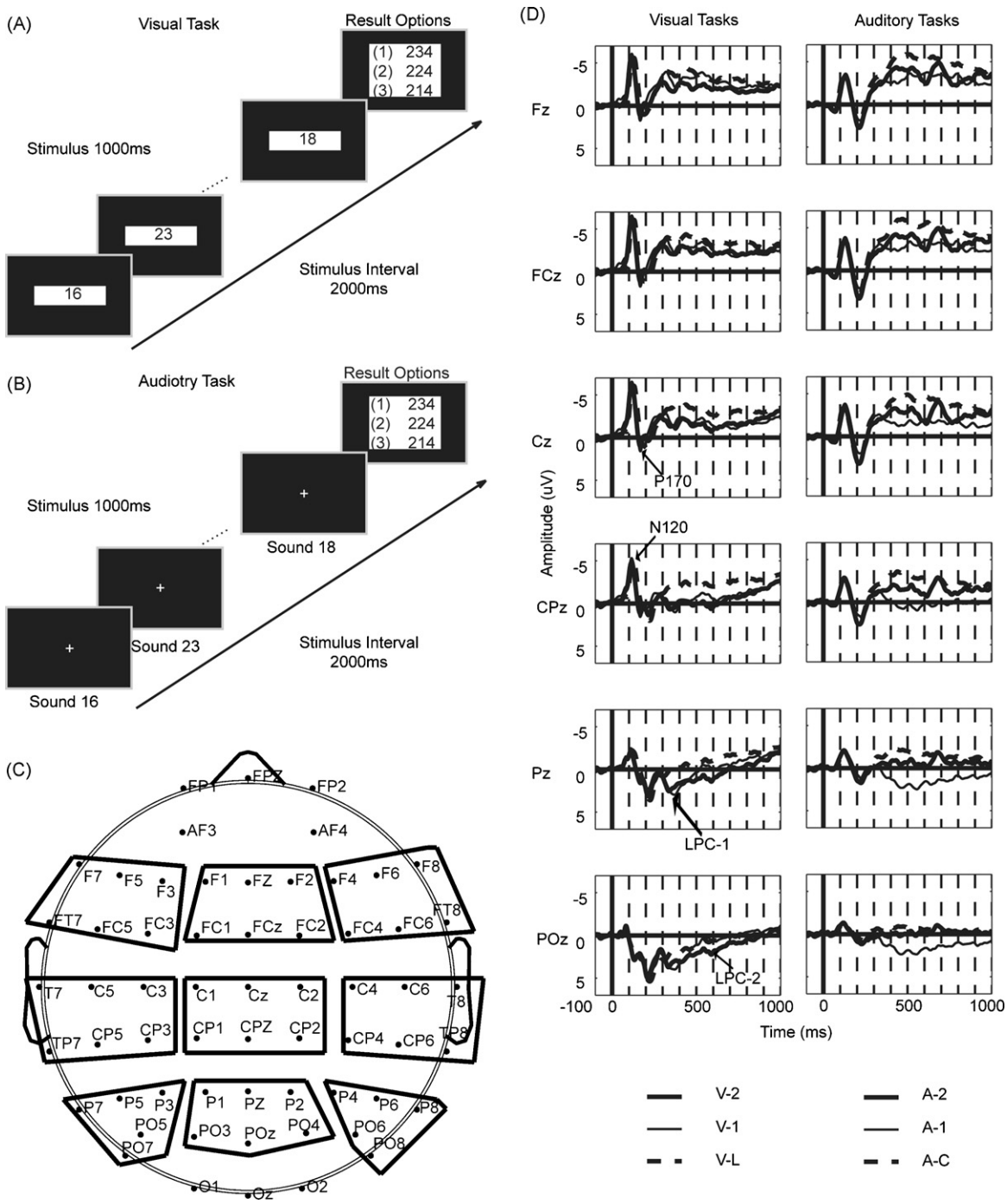

Fig. 1. (A and B) Demonstration of runs in visual and auditory experiments. In visual tasks, the visual stimuli could be 2-digit numbers, 1-digit numbers or letters. Likewise, in auditory tasks, the sound could also be 2-digit numbers, 1-digit numbers or letters. (C) EEG electrodes arrangement. The nine regions of interest (ROI) are displayed with the nine frames. (D) Grand average of ERP in visual and auditory tasks. The prominent peaks and troughs in visual tasks are named as routine.

while the other three (A-2, A-1, and A-L) were auditory tasks. Experiments are illustrated in Fig. 1A and B. Ten numbers (for mental addition) or 10 letters (for control tasks) were sequentially displayed in the center of the screen (for visual tasks) or played from the speakers beside the screen (for auditory tasks) every 2 s in each experimental run. Each stimulus was presented for 1 s. During the mental addition tasks, subjects were required to add the 10 numbers in succession and then select the result from one of the three options presented on the screen by pressing the buttons "1", "2", or "3" on a keyboard. V-2, V-1, and V-L presented 2-digit numbers (larger than 10 and less than 30), 1-digit numbers (from 1 to 9) and letters, respectively. Likewise, the auditory tasks were divided into three similar experimental groups. Subjects were instructed to watch the center of the screen in the visual tasks, while a cross was presented in the center of the screen during the auditory tasks in order to avoid extreme eye movements. During control tasks, subjects watched or listened to each letter in V-L and A-L, respectively, and randomly press buttons "1", "2", or "3" at the end of each run. Twenty runs were carried out for each task and were grouped into two blocks. The blocks of different tasks were presented randomly for each subject. Subjects took a 3–5-s rest between runs and rested 5–10 min between blocks depending on the individual. They were instructed to avoid eye movements during task performance.

EEG signals were recorded by an EEG recording system (SynAmps2, Compumedics, Ltd Corp, US). Sixty Ag–AgCl scalp electrodes (Quick-Caps, Neuroscan) were arranged in the standard 10–20 system (as shown in Fig. 1C). EEG signals from each electrode were referenced to linked earlobes. The impedance of each electrode was kept below 5 kΩ during experiments. Additionally, the electro-oculogram (EOG) signal was recorded to detect horizontal and vertical eye movements. EEG (60 electrodes) and EOG (2 bipolar electrodes) signals were filtered (0.1–100 Hz band-pass), amplified, digitized (1000 Hz sample rate), and stored for off-line analysis.

EOG contamination in EEG was excluded by means of linear regression implemented in Neuroscan EDIT software (Compumedics, US). Afterwards, the 60-channel EEG data were analyzed using Matlab 7.0 (Math Works, US) and EEGLAB 6.0 toolbox (Swartz Center for Computational Neurosciences, US) [6]. The time period from 100 ms before to 1000 ms after every stimulus onset was selected for ERP analysis. Ten sequential trials composed an exper-

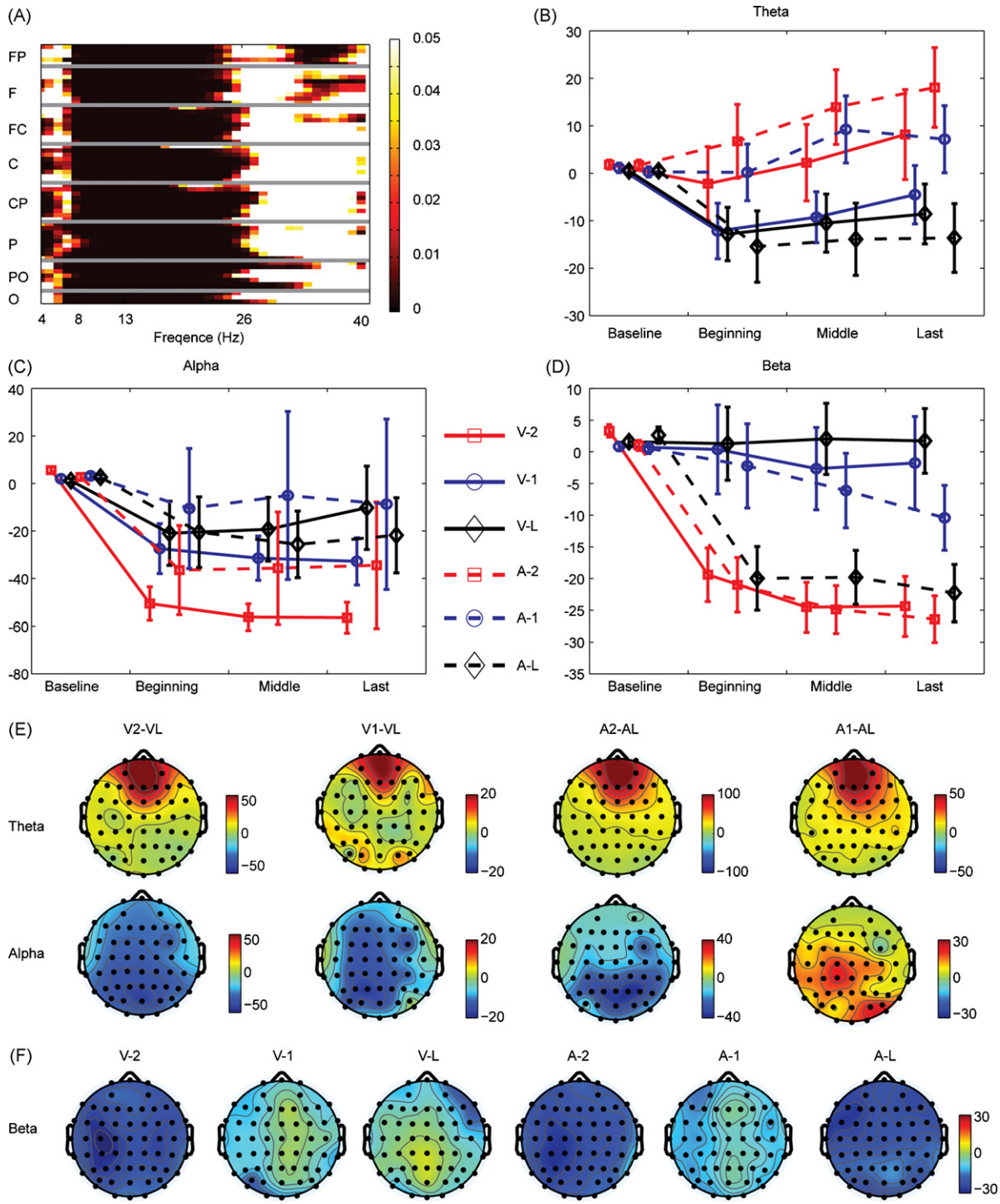

Fig. 2. (A) Point-by-point analysis of variance in Electrode-Frequency plane. The electrodes are arranged from front to back for rows, and from left to right for columns of each row. (B–D) Averaged event-related (de-)synchronization (ERD/ERS) values across subjects (mean ± SEM) at the Cz electrode in different frequency bands (theta, alpha, beta for B, C, D respectively). The baseline (2 s before the first stimulus onset of each trial) mean has been removed and scaled as 100-percent. (E) Differential topographies of theta ERS and alpha ERD values between additions and the control. (F) The topographies of beta ERD.

Table 1
Task performance: accuracy rate (AR) and reaction time (RT) (mean ± SEM).

	V-2	V-1	V-L	A-2	A-1	A-L
AR (%)	77.0 ± 2.5	98.0 ± 0.6		66.3 ± 3.8	96.3 ± 0.9	
RT (s)	1.95 ± 0.14	1.12 ± 0.07	0.92 ± 0.06	2.30 ± 0.18	1.32 ± 0.14	1.01 ± 0.18

imental run. In mental addition tasks, the trial following the first stimulus was excluded, as the calculation had not yet been performed. Only accurately answered runs were further analyzed. The joint probability of EEG voltages was computed for each trial at each electrode with trials containing values two deviations away from the mean defined as artifacts and excluded from further analysis. (The remaining trial number: V-2 132 ± 4; V-1 164 ± 3; V-L 166 ± 3; A-2 111 ± 6; A-1 162 ± 2; A-L 169 ± 2).

Event-related potentials (ERPs) were then calculated by averaging the remaining trials. The baseline was established by averaging the voltages from 100 ms preceding the onset of stimuli and was subtracted from the ERP voltages. ERP components were defined as peaks and troughs and named as routines (as in Fig. 1D). The amplitude of an ERP component was measured from its peak to the baseline (peak-value minus the baseline-value). The latency was measured from the stimulus onset to the peak.

Time-frequency analysis was used to determine the utilized frequency bands for ERD/ERS analysis. Short-time Fourier transform was applied to the data with a 250 ms Hanning window. Point-by-point analysis of variance was adopted from Molholm et al. [14] and applied to the spectrum values in different tasks at each electrode and each frequency point. The within-subject factor was TASK (V-2, V-1, V-L, A-2, A-1, and A-L). The spectrum values from 400 ms to 1800 ms after visual/auditory stimulus onset (early activities not included) were averaged for each corresponding frequency. The p values are shown in Fig. 2A in the Electrode-Frequency plane to find the significant frequency bands for ERD/ERS analysis.

All of the trials, following artifact removal, were then band-pass filtered into the frequency bands which showed significant difference in the ANOVA test, using a finite impulse response (FIR) filter with a 200-point Hanning window. The filtered values were thereafter converted into power by squaring. ERD/ERS were measured after averaging the envelopes and normalization with a baseline (2 s before the first stimulus of each run). ERD/ERS values from 400 ms to 1800 ms after visual/auditory stimulus onset were averaged for further statistical analysis.

Repeated measures analysis of variance (ANOVA) was used to compare performance (accuracy rate (AR) and reaction time (RT)), ERP components (amplitude and latency) and ERD/ERS values. Two-way repeated measures ANOVA was performed for RTs. The within-subject factors were MODALITY (visual and auditory) and DIFFICULTY (2-digit, 1-digit and control). Because the results were selected arbitrarily in the control tasks, the control groups were excluded from the AR analysis. Since the spoken numbers could not be presented as briefly as the visual ones, only visual ERP components were included for statistical analysis (four-way repeated measures ANOVA). The electrodes were grouped into nine regions of interest (ROI) as indicated in Fig. 1C, and ERPs from electrodes of each area were averaged to improve signal-to-noise-ratio. The within-subject factors were DIFFICULTY, AP (anterior, center and posterior), LR (left, medial and right) and SEQUENCE (beginning, middle and last). Nine calculation trials in each run were grouped into three sequential subsets which defined the three levels of the factor SEQUENCE. Five-way repeated measures ANOVA was performed for ERD/ERS values. The within-subject factors were MODALITY, DIFFICULTY, AP, LR and SEQUENCE (baseline, beginning, middle and last). All statistical analyses were performed with Statistica (StatSoft, US).

Task performance was averaged across subjects (Table 1). In control tasks, subjects made an arbitrary selection. Thus, the accuracy rates in the two control tasks were left blank in the table. Two-way repeated measures ANOVA showed a significant effect of MODALITY ($F(1,14) = 9.1$, $p < 0.01$) and DIFFICULTY ($F(1,14) = 94.2$, $p < 0.001$) for ARs. Significant interaction ($F(1,14) = 7.8$, $p < 0.05$) was found between the two factors. The post hoc Tukey HSD test revealed that AR in V-2 was significantly lower than those in V-1 ($p < 0.001$), and A-1 ($p < 0.001$), but higher than that in A-2 ($p < 0.01$). MODALITY ($F(1,14) = 9.3$, $p < 0.01$) and DIFFICULTY ($F(2,28) = 47.7$,

Table 2
ERP amplitude and latency (mean ± SEM) and the results of repeated measures analysis of variance (ANOVA) analysis. (NS) not significant.

	N120 (100–140 ms)	P170 (150–190 ms)	LPC1 (340–380 ms)	LPC2 (570–610 ms)
Latency				
Mean ± SEM (ms)				
V-2	120.1 ± 1.9	175.7 ± 1.6	354.2 ± 3.4	590.2 ± 1.1
V-1	121.7 ± 2.0	174.3 ± 1.7	369.0 ± 3.0	591.5 ± 2.0
V-L	124.9 ± 1.8	181.7 ± 1.6	354.3 ± 3.9	589.6 ± 1.2
Statistical results of ANOVA				
DIFFICULTY	$F(2,28) = 6.28^{**}$	$F(2,28) = 20.7^{***}$	$F(2,28) = 11.67^{***}$	NS
AP	NS	$F(2,28) = 12.6^{***}$	NS	NS
LR	$F(2,28) = 5.33^{*}$	NS	NS	NS
Amplitude				
Mean ± SEM (μV)				
V-2	-4.31 ± 0.44	2.37 ± 0.61	2.95 ± 0.49	1.84 ± 0.45
V-1	-4.47 ± 0.46	2.31 ± 0.55	2.79 ± 0.60	0.79 ± 0.65
V-L	-3.85 ± 0.42	1.76 ± 0.56	1.17 ± 0.62	-0.13 ± 0.47
Statistical results of ANOVA				
DIFFICULTY	NS	NS	$F(2,28) = 9.31^{***}$	$F(2,28) = 12.84^{***}$
AP	$F(2,28) = 54.87^{***}$	NS	$F(2,28) = 35.23^{***}$	$F(2,28) = 21.01^{***}$
LR	$F(2,28) = 21.68^{***}$	$F(2,28) = 6.05^{**}$	$F(2,28) = 4.76^{*}$	NS
SEQUENCE	NS	NS	NS	$F(2,28) = 7.64^{**}$

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 3
The results of repeated measures ANOVA analysis for the ERD/ERS values. (NS) not significant.

	Theta (4–8 Hz)	Alpha (8–13 Hz)	Beta (13–26 Hz)
MODALITY	NS	NS	$F(1,14) = 4.67^{*}$
DIFFICULTY	$F(2,28) = 3.35^{*}$	$F(2,28) = 9.85^{***}$	$F(2,28) = 8.37^{**}$
AP	NS	$F(2,28) = 8.1^{**}$	NS
LR	$F(2,28) = 12.12^{***}$	NS	$F(2,28) = 6.09^{**}$
SEQUENCE	$F(3,42) = 5.6^{**}$	$F(3,42) = 7.37^{***}$	$F(3,42) = 22.02^{***}$

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

$p < 0.001$) were also significant for RTs. No significant interaction was found between the two factors.

Averaged visual ERPs across subjects along the midline electrodes are shown in Fig. 1D. The mean values and the statistical results of the ERP components are shown in Table 2. N120 and P170 amplitudes were not significantly influenced by DIFFICULTY while the late positive components LPC1 and LPC2 showed significant changes with DIFFICULTY. The post hoc Tukey HSD test indicated that LPC1 amplitude in V-L was significantly smaller than those in V-2 ($p < 0.05$) and V-1 ($p < 0.05$), with no significant difference between V-2 and V-1. The LPC2 amplitude was larger in V-2 than those in V-1 ($p < 0.05$) and V-L ($p < 0.01$), with no significant difference between V-1 and V-L. SEQUENCE was only significant for the amplitude of LPC2 ($p < 0.01$) but not for the other three ERP components.

The statistical p values of the point-by-point (every point is from the Electrode-Frequency plane) ANOVA are shown in Fig. 2A and frequency bands with dominant significant p values ($p < 0.05$) across electrodes are selected for ERD/ERS analysis (theta: 4–8 Hz; alpha: 8–13 Hz; beta: 13–26 Hz).

Grand averaged mean values of ERD/ERS from the representative electrode Cz are displayed in Fig. 2B, C, and D for theta, alpha, and beta, respectively. The statistical results of ERD/ERS values are shown in Table 3. Theta ERS and alpha ERD values showed no significant difference between visual and auditory tasks. However, there were significant interactions between MODALITY and other factors. The post hoc Tukey HSD test demonstrated that the theta ERS was primarily located in the medial frontal areas and alpha ERD was predominant in the central and parietal areas. The differing topographies of the theta ERS and alpha ERD values between the addition and control groups are shown in Fig. 2E. The ERD/ERS values become larger with increasing task difficulty but the topographic patterns remain similar across modalities and difficulties (only Alpha ERD in A-1–A-L shows different patterns). Beta ERD was significantly affected by MODALITY, DIFFICULTY, and the interaction between them. The post hoc Tukey HSD test indicated that 2-digit tasks, similar to the auditory control task, generated larger beta ERD values than the other three tasks. The topographies of beta ERD in all the tasks are shown in Fig. 2F.

Reaction time and accuracy rate were significantly influenced by both MODALITY and DIFFICULTY. Difficult 2-digit tasks resulted in decreased accuracy rates compared to the 1-digit tasks, in which accuracy rates were approximately 100-percent.

Early ERP components, which might be stimulus-driven, are typically phase-locked to the stimulus onset and stable across subjects. Therefore, the ERP average would not eliminate them and thus could be used to indicate the sequential processes during different kinds of cognitive tasks [12]. N120 and P170 are regarded as ERP components enhanced by visuospatial discrimination [9]. The late positive slow waves are considered as calculation components [9]. We observed dissociable late components LPC1 (around 360 ms) and LPC2 (around 590 ms) as we manipulated the calculation difficulty and added 2-digit addition tasks. LPC1 was more prominent in calculation tasks than in the control task and LPC2 was most prominent during 2-digit addition compared to the other two tasks. This might indicate that LPC1 is generally evoked during mental calculation and LPC2 is primarily elicited for number-size effect and in more difficult tasks. LPC2 was also significantly affected by SEQUENCE and became larger in the later trials of each run, which further suggested the relationship between LPC2 and task difficulty. Similar ERP waveforms to our results were reported by Pauli et al. [18] who found that the offset of a positive slow potential was prolonged in proportion to difficulty in a multiplication task.

Event-related (de-)synchronization analysis is suitable for detecting oscillatory changes that are not ideally phase-locked to the stimuli. Studies showed that theta ERS usually coexisted with increasing task difficulty and were correlated with cognitive functions such as working memory [10], language [21], and attention [1]. Here we also observed increasing frontal midline theta power with increasing task difficulty, which was independent of stimulus modality. The amplitude of the alpha band activity is suppressed by eye opening, and visual stimuli, whereas it is enhanced during internal tasks such as mental calculation and working memory [17]. It was thought to reflect idling or inhibition of task-irrelevant cortical areas. However, recent research has suggested a direct and active role for the alpha oscillation in the mechanisms of attention and consciousness [17]. We also observed decreased alpha activity in parietal areas, independent of modality, during mental addition. The frontal theta and the parietal alpha together support Zago et al.'s finding [25] and give additional information in frequency domain which is important for the communication between brain areas [11].

Oscillation in the beta band has long been thought to represent sensorimotor processing [15,19]. However, some studies have suggested that beta oscillations may also play a role in working memory, attention [23], and emotional process [20]. Desynchronization in the 10–20 Hz frequency range was reported during recognition [7]. Interestingly, similar beta ERDs were found in our study for 2-digit additions (both visual and auditory) and auditory control, but distinct from those in 1-digit addition tasks. This result, combined with the LPC2 discrepancy between V-2 and V-1, indicated differential patterns within the calculation group. The increasing calculation difficulty might lead to the involvement of different brain networks, which might be auditory-related resources, instead of visuospatial networks. Dehaene et al. reported that exact and approximate arithmetic involved disparate brain systems and claimed that exact arithmetic was acquired in language-specific format, while approximate arithmetic recruited bilateral areas of the parietal lobes involved in visuospatial processing [5]. The phonological loop, one of the working memory subsystems [2], which was activated by auditory input, was found to contribute to the calculation process [25], and in this case the phonological loop might be the candidate of neural substrate.

In conclusion, our results dissociate the computation components into two different parts. One is connected with calculation generally and the other increases with calculation difficulty. Similar oscillatory patterns in the theta and alpha frequency bands are found for both the visual and auditory addition tasks, which may indicate similar oscillatory brain networks involved in the two modalities, leading to an amodal calculation system for normal people, which has already been reported for abacus mental calculation experts [3]. Our results also showed similar beta ERD in 2-digit additions and the auditory control, which may reveal some connection between the phonologic loop and calculation. This outcome coincides with the theory of language-specific format in exact arithmetic and visuospatial processing for approximate arithmetic [5]. The fronto-parietal network is activated in both working memory [16] and calculation [25], which may provide the neural substance for these results. Further interruption investigation (such as tran-

scranial magnetic stimulation (TMS)) needs to be done for more conclusive evidence.

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